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RESEARCH NOTE

BUILT TO LEARN: HOW WORK PRACTICES AFFECT EMPLOYEE LEARNING DURING HEALTHCARE INFORMATION TECHNOLOGY IMPLEMENTATION[1]

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We test the hypothesis that work practices complement IT investment, in part, by accelerating how rapidly employees acquire the skills needed to use new IT systems. We combine support request data from an EMR vendor with survey responses on work practices from 962 employees from 15 client nursing homes. Nursing homes using work practices that prior studies have shown to complement IT investment—those promoting discretion, teamwork, training, high staffing levels, and communication—experienced more rapid declines in requests for technical support. We then show that these benefits are due, in part, because the use of these work practices facilitates learning for workers in frontline occupations who otherwise may not have the freedom to experiment with and adapt the new technology systems. For many frontline workers, discretion was more important than training in explaining IT learning. Implications for the healthcare industry are discussed.

Keywords: Healthcare information technology, IT implementation, electronic medical records, organizational complements, nursing homes, EMR, learning, software-as-a-service

Introduction

A defining feature of the modern information systems environment is the cross-boundary nature of IS delivery, as seen, for example, in software-as-a-service (SaaS) or IT outsourcing agreements (Hirschheim et al. 2002; Lacity et al. 2010). Much of the literature on this model of delivery focuses on variation in vendor practices, contract design, and technology in driving IT outcomes (see, for example, Barthélemy and Geyer 2005; Lee and Kim 1999; Levina and Ross 2003). Yet, despite the fact that these factors are often the same—many packaged IT systems are installed by the same vendors using common installation practices and standardized contracting arrangements—there is still considerable outcome variation.

One factor that may be central to this variation is client learning. Learning capabilities are an important predictor of firms' adaptation to new technologies (Argote and Miron-Spektor 2011; Kogut and Zander 1992; Pisano et al. 2001). Recent literature links learning to important IS development outcomes (Boh et al. 2007; Singh et al. 2011), implementation (Wu et al. 2012), use (Kane and Alavi 2007; Mukhopadhyay et al. 2011), and support (Kim et al. 2012).[2] At client sites, learning is likely to be influenced by the ability to assimilate new knowledge, which can be heterogeneous across firms (e.g., Argote and Ingram 2000), and prior work documents a

[1]Ravi Bapna was the accepting senior editor for this paper. Gautam Ray served as the associate editor.

[2]Also see a review of the prior literature in Robey et al. 2000.

relationship between work practices and the diffusion of employee knowledge (Preuss 2003).

In this study, we test the hypothesis that IT-complementary work practices accelerate workforce learning related to new IT implementation. Existing literature documents the importance of organizational complements for explaining IT returns (e.g., Bresnahan et al. 2002), but little attention has been paid to how these complementarities arise. To the best of our knowledge, this study is among the first to argue that work practices complement IT investment, in part, by facilitating workforce learning (a recent exception is Tambe et al. 2017).

Theoretically, work practices that promote employee involvement, discretion (defined as the extent to which employees are provided with professional autonomy and decision making authority), teamwork, training, and staffing are likely to affect the acquisition and transfer of new knowledge within the organization, thereby influencing IT learning. IT complementary work practices may enhance skill development through superior training and staffing practices and by allowing more employee discretion, which enables employees to learn through trial-and-error (Batt 2002; Osterman 1994). Moreover, use of these practices encourages sharing of new technical knowledge through communication and the use of self-managed teams. Greater reliance on teams has been shown to increase interaction, which promotes learning across members, and the diffusion of individually acquired technology skills. This linkage is supported by HR management research, which has provided evidence for a link between work practices and knowledge generation (e.g, Huselid 1995; Preuss 2003).

We also examine how the relationship between work practices and IT learning differs across occupational groups. Prior work on IT skill formation focuses on jobs characterized by high levels of discretion and teamwork (for an examination of IT services, see Bapna et al. 2013). Nevertheless, many settings—healthcare is a prominent example—differ in their job structure, both in terms of their work environment and because they are characterized by heterogeneity in skill levels, discretion, employee involvement, and training across occupations (Weinberg et al. 2011). For instance, in nursing homes, registered nurses (RNs) are employed in supervisory positions characterized by higher discretion. Certified nursing aides (CNAs), on the other hand, are assigned to the vast majority of frontline, direct resident care-related tasks, and tend to be less skilled (Eaton 2000). We test the hypothesis that the benefits of these work practices arise, in part, because their use facilitates learning for workers in frontline occupations who may not have otherwise had the flexibility to experiment with the IT system and adapt it to their workflow.

This study presents findings from the analysis of a data source generated by a program evaluation conducted in the state of New York in which a single vendor was selected to provide electronic medical records (EMR) systems (both hardware and software) to 15 client nursing homes from 2007–2008. Nursing homes were chosen to minimize heterogeneity along dimensions such as resident mix, so the research setting offered an opportunity to study EMR implementation in a relatively controlled setting. Measures of workplace organization at nursing homes, collected immediately prior to implementation, were developed through surveys of 962 employees from the in-sample nursing homes. The dependent variable was constructed from the time-stamped contents of a support request database maintained by the vendor. These support requests provide a proxy measure for the extent to which nursing home employees required support while using the EMR system. An advantage of this database is that it distinguishes support requests related to system bugs, which should not be affected by employee learning, from support requests related to end users' proficiency with the new system.

The design of this study has caveats: most significantly, it is restricted to a small number of nursing homes. The evidence we provide is also correlational; the causal impact of implementing these practices on employee learning outcomes is beyond the scope of our analysis. In exchange, however, it offers two advantages that are difficult to replicate with larger, archival datasets. First, because the participants were, as a part of the pilot program design, selected to receive the EMR system from a larger pool of nursing homes, our results are somewhat less subject to criticisms based on selection effects that could arise from only observing adopting organizations. Potentially confounding influences, such as union membership, geography, staffing, and wages, which might otherwise affect our estimates are, by design, similar across the facilities in our sample, making direct comparisons across facilities more meaningful. Moreover, the EMR system, vendor, implementation, and support processes are the same across installations, mitigating concerns about implementation differences. Second, the dependent variable is a relatively fine-grained measure of the learning required by employees during the implementation of a new EMR system. This granularity of measurement would be difficult to replicate for heterogeneous samples of firms.[3]

[3]Our research design draws from literature that argues that assembling detailed operational and outcome data from a specific, controlled production process is useful for isolating the effects of new IT on organizational outcomes (see Bartel et al. 2007; Litwin 2010). It is similar in its approach to recent work in the IS literature that uses fine-grained measures of information use generated by frequent interaction with digital tools, often from within a single firm, to reveal relationships between IT and worker productivity (e.g., Aral et al. 2012).

We find that end-user learning requests decline more rapidly in nursing homes that use work practices that the prior literature has shown to complement IT investment. We also find that the efficacy of these practices varies across occupational groups. For low-skill workers, discretion appears to have a more pronounced effect on technical skill formation compared to training. Although training has the expected effects on high-skill occupations, differences in the levels of discretion that managers provide to workers is more important for IT learning for workers in lower-skill occupations.

Our findings contribute to the literature on IT-related complements. No prior work, to the best of our knowledge, demonstrates that IT-complementary work practices affect IT outcomes by accelerating skill development among workers. Our study also contributes to the healthcare IT adoption literature. Recent work documents the impact of EMR systems on performance and medical outcomes (Agha 2014; Bhargava and Mishra 2014; Dranove et al. 2014; Hitt and Tambe 2016; Jha et al. 2009). Despite the allocation of a tremendous level of investment toward overhauling the U.S. healthcare system's IT infrastructure, the evidence on outcomes has been mixed (see Avgar, Litwin, and Pronovost 2012). Our study provides one explanation for this variation. The organizational differences we study may be particularly salient in the healthcare setting. Given the high level of interdependence across units and occupational groups (Gittell et al. 2010) along with the need to address multiple, competing stakeholder needs and pressures (Avgar, Givan, and Liu 2011), variation in work practices is likely to be especially pronounced in healthcare.

Our findings also have implications for policy and practice. They suggest that the diffusion of health IT requires that, alongside investments in the technology, policy makers allocate resources to the establishment of adequate organizational infrastructure capable of facilitating IT learning. Therefore, in an EMR setting, training is likely ineffective in the absence of adequate levels of discretion provided to low skilled frontline healthcare workers. Hospital administrators and managers should, therefore, ensure that efforts to promote employee discretion and involvement are not solely aimed at high skilled healthcare professionals.

Case Examples from Two Nursing Homes

To motivate the empirical analysis we present below, we provide contrasting examples from field research in two different nursing homes. *Senior Care*[4] was representative of an organization employing IT complementary practices. Even prior to the EMR implementation, Senior Care administrators viewed employee involvement, communication, and discretion as central components of their managerial strategy. Senior Care was also more advanced in its use of teams than other nursing homes in our sample, and they adopted a cooperative labor-relations approach characterized by high levels of trust and a strong relational foundation with the union. This dynamic was aligned with Senior Care's efforts to engage their workforce and provide them with participation opportunities. Finally, Senior Care was advanced in delivering care consistent with residents' needs, known as resident-centered care ("patient-centered care" in the hospital setting; for a review, see Avgar, Givan, and Liu 2011).

In our qualitative analysis, we observed a link between the use of these work practices and core elements of the implementation trajectory. Frontline workers were given the latitude to experiment with the technology and to find effective ways to use it, which supports the trial-and-error learning argument above. Furthermore, Senior Care identified technology "champions" among frontline staff and empowered them to assist others with use of the new technology. Finally, Senior Care viewed the introduction of new IT as a vehicle through which to reinforce its discretion and involvement-centered practices and employees viewed the technology as a tool that provided them with the capacity to provide better resident care. In sum, work practices that promoted discretion, teamwork, and collaboration influenced how frontline employees engaged with the technology, affecting the acquisition and diffusion of IT related knowledge.

Elder Care was representative of a different approach, one that was hierarchical and was not characterized by discretion and involvement-centered practices. Work related decisions were made in a top-down, authoritative manner with no focus on frontline input. Elder Care administrators had few mechanisms in place to facilitate employee voice, and communication between top management and frontline staff was infrequent. Elder Care's organizational approach carried over to its labor-relations dynamic, which was characterized by adversarial union relationships. Elder Care took a traditional, institution-centered approach to care delivery.

These organizational practices, which were in place prior to the introduction of new IT, were linked to its IT implementation. Elder Care's administrators focused on the potential of the technology to monitor and control their workforce. IT was viewed as a way to complement other tools such as surveillance cameras and compliance-centered managerial practices. Unlike Senior Care, Elder Care did not have work practices that facilitated frontline staff engagement with the new system and fostered experimentation and learning.

[4]To protect anonymity, we have changed the names of the two nursing homes we review to Senior Care and Elder Care.

Interviews with frontline employees at Elder Care documented limited discretion around the implementation of the technology. In addition, the organization's focus on the control and monitoring potential of the new technology created resistance on the part of frontline employees. As a result, Elder Care employees were less attuned to the link between the new technology and resident care, which translated into decreased incentives to learn its effective use (for a discussion of the relationship between managers' views of new technology and the perceptions of frontline users during implementation, see Edmondson et al. 2001). Finally, Elder Care did not empower frontline technology champions to assist in the implementation process. These relationships underscore the link between work practices and conditions for frontline IT learning.

Data and Methodology

Data

The data for this paper were collected as part of a larger demonstration project in the state of New York that funded the introduction of EMR technology in New York City area nursing homes. For our study, we examined the implementation of an EMR system provided by a single provider in 15 pilot organizations, chosen by the Quality Care Oversight Committee (QCOC) from approximately 150 candidates in the region to minimize differences in workforce training, basic preparedness, and patient population.[5] The vendor used a standardized implementation process, which means that upfront technology, installation, and implementation costs are similar across nursing homes, except for some small and predictable variation due to size.

Implementation of the new technology took place over a period of approximately 12 months with staged installations across organizations, which allows us to capture variation in the vendor's implementation experience. The system was delivered as a web application through a central data center, which is also a largely fixed cost with a small per-patient (resident) variable cost that is essentially identical across nursing homes. The delivery of the technology through a web application also ensured that all nursing homes were running the same version of the system at any point in time. What varied more significantly across nursing homes was the degree of post-implementation support required. This support was principally provided through telephone contact with vendor personnel, and was provided at no charge other than the standard per-resident monthly fee charged to the nursing homes by the vendor for ongoing use of the system. To test our hypothesis, we use the following new data sources generated by this research context.

Employee Surveys

Variation in employers' work practices was collected through employee-level surveys. Surveys were administered to frontline healthcare workers in the nursing homes, immediately prior to receiving the EMR system. Survey data were collected between June 2007 and July 2008 by phone through the Cornell Survey Research Institute. We included in the sample all frontline staff from the participating organizations that fell into three occupational categories: (1) nurses (registered nurses and licensed practical nurses (NRS); (2) certified nursing assistants (CNA); and (3) allied professionals which includes social workers and other support occupations (ALP). These occupational groups represent the primary users of the various EMR system components. In total, we contacted just over 2,000 employees across the 15 organizations. Of these, 962 completed the survey for a response rate of approximately 48%.

Support Requests

Our second data source was a database provided by the EMR vendor. To handle support requests after health information technology (HIT) implementation, the vendor used a request tracking system to record when nursing homes issue support requests after installation. Quantities of support requests indicate general support levels required by individual nursing homes after the initial implementation, and are a measure of the organization's facility with the new technology. We use 13 months of support request data in our analysis.

These support request data include issue descriptions, the time and date at which the ticket was issued, as well as whether the ticket is related to a technical system malfunction or related to an issue that reflected the end-user learning process. Specific examples of the descriptions associated with some representative support requests include (1) requests about how to use features of the software system (end-user learning) and (2) bug reports associated with malfunctioning software or hardware (system bugs). Table 1 provides examples of the former support requests, which are the principal focus of our study. In distinguishing between the two types of support requests, we principally relied on the vendor's categorization, which was put in place in an effort to effectively deal with different types of IT learning challenges. However, to ensure that our results were not sensitive to the vendor categoriza-

[5]While the QCOC chose recipient nursing homes in order to reduce training and patient population differences, there was an effort to ensure a diverse sample of organizations in terms of size, geographic location, and ownership.

Table 1. Examples of Support Requests

"Adriana has MDS & Care Plan questions about interventions and Rap notes."
"User needs to change the assessment type for one of the MDS assessments."
"Would like to know how transfer physician orders from one duplicate resident to another"
"Cannot find a specific order on a resident even though it is showing up on the documentation history."
"Needs to change her ADT info"
"Needs to Edit Care Plan Evaluation Notes for a resident"
"Progress note entered for duplicate resident; how to move it?"
"How do I Add A Bed Hold for a Discharged Resident?"
"How to change the completion date of an assessment?"
"How do I put an Order on Hold for Resident System ID 104151?"

tion, we had two independent research assistants hand-code all of the support requests into the two categories, and reran our analyses. These robustness results are omitted here for brevity but are available upon request.[6]

Support requests specific to user learning are a unique measure of a particular source of disruption that takes place in organizations during IT implementation. Much prior research views learning through a lens of improved efficiency (Argote 1999), but there has been a significant broadening of scope in measures that effectively capture learning (for a review of studies that have examined broader measures, see Argote and Miron-Spektor 2011; Lapre and Nembhard 2011). We view support requests as a measure of the extent to which frontline staff have mastered or learned the skills needed for effective EMR use.

Supplementary Data

Supplementary data on individual nursing home characteristics, such as the number of beds and various quality indicators, were collected through the "Nursing Home Compare" dataset available through the U.S. Department of Health & Human Services.

In addition to these archival survey data sources, the authors participated in numerous meetings, interviews, observations, and discussions with managers and frontline employees from each nursing home, with vendor personnel, and with the committee responsible for the execution of the demonstration project. This field experience was instrumental when operationalizing our constructs. For example, our discussions with the implementation vendor provided us with an in-depth understanding of the manner in which support requests were categorized.

Variables

Dependent Variable

Our dependent variable is the number of support requests issued in each month for each nursing home. We aggregated support requests to the month because it eliminates the noise that appears at lower levels of aggregation. For some nursing homes, this generates measurement error associated with partial start-up months, so in our analyses, we report the results of tests that exclude the first month of use for each nursing home. In most of our regressions, we focus on support requests related to end-user learning, although in some regressions, we report results from support requests related to system bugs as a robustness test.

One limitation of using this dependent variable is that we cannot directly distinguish from the data whether declining support requests reflect greater familiarity with the EMR system or falling use. However, the *% HIT use in home variable* measure, described further below, indicates that during the sample period, EMR use grew steadily in the nursing homes. Interviews with both nursing home managers and the EMR vendor supported this observation.

Independent Variables

Work practices (ORG). We measure work practices (ORG) using the perceptions of employees in different roles at the

[6]For these supplementary analyses, we only retained support requests in the database upon which the RAs agreed with one another, and we discarded support requests that the RAs both coded as duplicate requests. The regression results using this recategorization are available upon request and are qualitatively similar to the regression results reported in the main text.

nursing home (e.g., Bowen and Ostroff 2004). Research has differentiated human resource management (HRM) practices that are rated by a single HR manager from employee perceptions of organizations' HRM practices, which can vary across individuals depending on a range of personal factors including individual frame of reference (Den Hartog et al. 2004; Kuvaas 2008). Prior literature argues that employee perceptions of HR practices exert influence on how employees feel and behave at work (Alfes et al. 2012; Edgar and Geare, 2005).

Our ORG measure included five practice dimensions. The literature has used differing but closely overlapping measures of IT-complementary workplace organization focused on employee discretion, teamwork, job quality, and training.[7] The five practices we measure are (1) teamwork and (2) decentralized decision-making (six items with $\alpha = .75$, e.g., "Employees in my unit have a great deal of say over how work is done" and "When a problem arises in my unit, we generally try to solve it as a group"), (3) training (two items with $\alpha = .70$, e.g., "My nursing home places a strong emphasis on training"), (4) communication (five items with $\alpha = .72$, e.g., "Employees in my unit share information freely with each other"), and (5) staffing (two items with $\alpha = .73$, e.g., "My nursing home has sufficient staff to provide good quality of care to residents").

For all items, we used a five-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*). We took the standardized sum of the five sub-practices to construct a single ORG measure from each set of employee responses, based on prior research (e.g., Bae and Lawler 2000). Cronbach's alpha for this ORG measure was .85 with an inter-rater agreement (rWG) of .83. Finally, ORG measures at the nursing home level were created by averaging the measures created from the responses of the employees within each nursing home. Our construction of this organizational practice variable, using a standardized sum of several related survey-based constructs, is based on prior work that constructs work practice based measures from organizational surveys (Breshanan et al. 2002; Huselid 1995; Osterman 1994; Tambe et al. 2012).

[7]For instance, Osterman (1994) considers quality circles, job rotation, teamwork, and total quality management. Batt (1998-1999) measures employee discretion and empowerment, quality circles, training, self-regulation and coordination in teams, and perceptions of job security and possibility for promotion. Black and Lynch (1996) consider employee voice, profit sharing, recruitment strategies, and use of high-performance work systems including total quality management, benchmarking, number of employees per supervisor, self-managed teams, and employee training. Bresnahan et al. (2002) measure use of self-managed teams, employee involvement, team building, promotion based on teamwork, job breadth, and discretion about the pace and method of work.

Nursing Home Experience: Our measure of the amount of experience ($NHExp_{it}$) a facility (i) has with the EMR system prior to a given month (t) is the logged cumulative number of beds serviced by the new technology through month t-1. This measure is computed using the product of the number of beds in the nursing home and a measure indicating the percentage of care orders executed with the HIT system *(% HIT use in home*), which is described below.

Vendor Experience: Vendor experience is measured as the logged number of total beds served by the vendor across all nursing homes, not including the current home-month. It is important to note that this was a new vendor, and prior to the demonstration project, the vendor had installed this HIT technology in only one nursing home not included in our study. Therefore, our vendor learning measure captures almost all of the vendor's experience at the point the study began.

% HIT use in home: Levels of HIT use are measured as the percentage of total care orders processed using the HIT system. This variable captures the degree to which frontline nursing home employees are actively using the new system (as opposed to simply having installed the system). Usage measures are important because prior research has shown the importance of usage levels in understanding the impact of new IT systems, especially in the healthcare industry (Devaraj and Kohli 2001). We include HIT use levels to distinguish between nursing homes that have a small number of support requests and are successfully using the system without problems from those that have a small number of requests because they are not actively using the system.

CNA Minutes: We include the logarithm of the number of *certified nursing assistants (CNA) minutes per resident per day*, to capture differences in staffing levels across nursing homes. We measure CNA minutes because these are the primary frontline caregivers in the nursing home setting with responsibility for a large proportion of the documentation of resident care using HIT.

Beds: As a measure of size, we include the logarithm of the *number of beds*, because smaller nursing homes may have a systematically easier or more difficult time implementing new IT systems, and because facility size might affect the volume of service requests. Number of beds is the most common measure of both size and productive capacity in an extensive literature on healthcare and nursing home economics (e.g., see Bostick et al. 2006; Cohen and Spector 1996).

Nonprofit: To identify nonprofit nursing homes in the sample, we include a dummy variable that takes the value 1 when a nursing home has nonprofit status. The nursing home industry is comprised of both for-profit and not-for-profit institutions.

Table 2. Summary Statistics for Key Variables

	Mean	Std. Dev.	Min	Max
Support requests (per month)	24.3	16.8	1	89
Client-side requests (per month)	8.1	6.6	1	34
Vendor-side requests (per month)	15.7	12.3	1	61
Support requests per bed (per month)	.115	.074	.0037	.3667
Number of beds in nursing home	215.2	56.5	120	320
CNA minutes per resident	137.9	27.7	94	209
EMR system utilization level	87%	31%	0.00%	99.9%
ORG measure	0	1	-0.308	0.498

Descriptive Statistics and Correlations

In Table 2, we report summary statistics and extreme values for key variables. The average number of support requests per month issued by each of the nursing homes in our sample is just over 24, an average of about 1 request per month for every 8 or 9 beds in the nursing home. Of these, just over 15 per month are support requests related to system bugs and about 8 per month are requests related to end-user learning, with the remainder falling into smaller categories such as requests to return equipment. The average number of beds in our sample is about 215. This is substantially larger than the average US nursing home, which has about 105 beds.[8]

Figures 1a and 1b illustrate the variation in requests that occurred across nursing homes. Each figure plots the trajectory of support requests for the sample, where nursing homes are grouped according to whether they fall above or below the median ORG value for the sample. The trend lines are smoothed local polynomial regressions based on the individual nursing home-month observations. Figure 1a illustrates the trajectories related to system bugs, which should not be expected to differ by the organizational characteristics of the client site, and indeed, the trajectories appear similar across groups of nursing homes. By contrast, a comparison of the trend lines in Figure 1b suggests that support requests related to end-user learning declined more rapidly for high ORG firms.[9] This variation is interesting given the other similarities across organizations noted above, and is consistent with the key hypothesis of this paper, which is that workers in nursing homes using IT-complementary practices more quickly gained the skills needed to use the EMR system. The next section tests these relationships in a regression format.

Regression Model

We model the decline in support requests using a count model, and we include interaction terms to test whether workplace organization explains learning variation. We estimate the following exponential model, which prevents negative values.[10]

$$\Pr(Y = y) = \frac{e^{-y}\lambda^{y}}{y!} \tag{1}$$

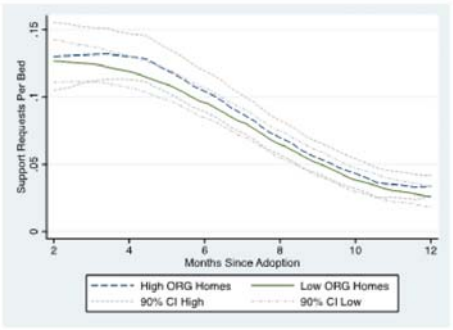
$$E(Y) = \lambda = e^{xB} \tag{2}$$

In this model, λ is the Poisson parameter and Y is the count variable to be explained. The Poisson model forces the mean and variance to be equal. For data sets such as ours, where the variance is larger than the mean, negative binomial models are more appropriate (Hausman et al. 1984). The negative binomial model is a generalized version of the Poisson model that allows for unobserved heterogeneity in the probability of occurrence between otherwise identical observations. In a negative binomial model, the dependent count variable λ is specified as $exp(x\beta + \varepsilon)$ where ε is a gamma distribution and where x is a vector of covariates, which produces a negative binomial distribution for the number of occurrences (Kennedy 2003).

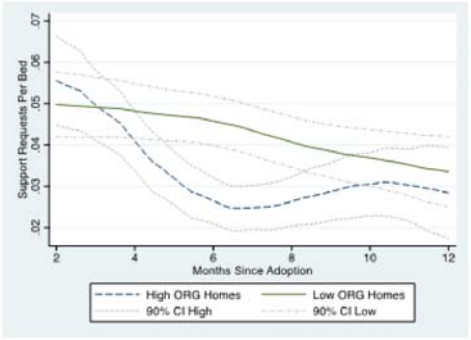
[8]Average nursing home size numbers are taken from the CDC (http://www.cdc.gov/nchs/fastats/nursingh.htm).

[9]Like most packaged software implementations, the majority of the site-specific customization is in the form of "configuration" in which parameters that control the operation of the system are entered into a database. With the exception of some interfaces to other systems in the nursing home (e.g., a pharmacy system interface), the base software is identical across nursing homes. As a result, variation across nursing homes in technical support calls is driven by how well the software package was implemented and not the overall quality of the underlying package.

[10]OLS models are poorly suited for analyzing raw count data because count data are nonnegative and nonnormal. Our analyses, which use support requests as the dependent variable, are based on negative binomial regressions, which are commonly used to analyze count data.



(a) Decline in System Bugs Per Bed



(b) Decline in End-User Learning Requests Per Bed

Figure 1. Monthly Declines in Support Requests Per Bed

To test for experience-based learning, we estimate negative binomial models where λ, the number of support requests, is associated with the following covariates:

$$ln(\lambda)_{it} = \beta_1 NHexp_{it} + \beta_2 ORG_{it} + \beta_3 (ORG_{it} * NHexp_{it}) + controls + month_t + facility_i + u_{it} \quad (3)$$

In this model, *NHexp* represents total nursing home experience with the technology, and *ORG* is a measure of the organizational practices used at the nursing home. A negative and significant estimate on the nursing home experience term (β_1) indicates that support requests decline with experience. A significant coefficient estimate on the ORG main effect indicates that the use of IT-complementary work practices is statistically associated with differences in overall support requests issued by the nursing homes. A significant and negative coefficient estimate on the interaction term between ORG and experience is consistent with higher ORG firms benefitting more rapidly from experience. As a robustness check, some of our analyses use requests related to system bugs, rather than end-user learning, as the dependent variable. If variation in requests is due to work practices that facilitate learning, we should only observe our results for support requests relating to end-user learning. Greater numbers of system bugs should not be affected by work practices, and therefore provide a useful robustness check for our results.

The model in equation (3) also includes control variables for HIT usage, size (beds), not-for-profit status, staffing levels measured as CNA minutes per resident per day, and a measure of vendor experience with the technology, and it includes fixed-effects for month and nursing home. Each of the control variables can affect the rate of learning. Larger organizations face additional organizational complexities, which can increase numbers of problems. Utilization rates are also likely to affect the rate of requests issued. Finally, CNA minutes per resident per day proxy for staffing adequacy, which may affect organizational outcomes (Avgar, Pandey, and Kwon 2012). We use a measure of CNA minutes since CNAs are the primary frontline caregivers in the nursing home setting and therefore are the primary source of HIT documentation. Finally, the distinction between for-profit and not-for-profit institutions has implications for the mission of the organization, which affects strategies and structures.

In our regressions, we utilize robust standard errors, clustered by nursing home to account for correlations in errors among observations for the same facility in different months. In many of our regressions, we augment this model by including fixed-effects at the nursing home level. STATA estimates these models using maximum likelihood methods implemented in the procedure NBREG. In our analysis, we also report results using the Poisson estimator that is built into STATA because of the known consistency issues with the negative binomial estimator (Allison and Waterman 2002; Simcoe 2008; Wooldridge 1997).[11]

Results

Main Results

All regressions in Table 3 use support requests as a dependent variable and include dummy variables for each month. In column (1) of Table 3, we find that greater levels of nursing home experience are associated with fewer support requests, which is consistent with learning-by-doing. The coefficient estimate on the ORG measure in equation (2) is negative and

[11]We thank an anonymous reviewer for suggesting this test.

Table 3. Aggregate Organizational Measure (ORG) and End-User Support Requests

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Support Request Type	**End-User Learning**	**End-User Learning**	**End-User Learning**	**System Bug**	**End-User Learning**	**System Bug**	**End-User Learning**
Model	Neg. Bin.	Neg. Bin.	Neg. Bin.	Neg. Bin.	Neg. Bin.	Neg. Bin.	Poisson
NH experience	-0.076***	-0.074**	-0.071***	0.064	-0.011	0.053	-0.016
	(0.029)	(0.031)	(0.025)	(0.046)	(0.028)	(0.056)	(0.026)
ORG		-0.262***	-0.044	0.077	-0.192	-0.121	-0.148
		(0.068)	(0.103)	(0.210)	(0.121)	(0.182)	(0.120)
ORG x NH exp			-0.034**	-0.015	-0.036**	-0.020	-0.046***
			(0.017)	(0.027)	(0.015)	(0.023)	(0.014)
HIT Use (%)	0.476**	0.703***	0.566***	0.164	0.648***	0.560*	0.597***
	(0.214)	(0.210)	(0.198)	(0.280)	(0.192)	(0.336)	(0.199)
Vendor experience	-18.70***	-17.36**	-17.16**	0.087***	-14.24**	0.050	-17.04**
	(6.863)	(7.031)	(7.126)	(0.031)	(6.824)	(0.035)	(7.156)
Log (CNA Minutes)	0.126	0.322	0.348	-0.263			
	(0.535)	(0.415)	(0.417)	(0.237)			
Log (Beds)	0.526	0.311	0.273	0.254			
	(0.374)	(0.234)	(0.229)	(0.169)			
Non-Profit	-1.147***	-1.485***	-1.452***	-0.675***			
	(0.0989)	(0.160)	(0.154)	(0.101)			
Month effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Facility effects	No	No	No	No	Yes	Yes	Yes
Observations	185	185	185	196	195	195	207

Notes: Robust standard errors are shown in parentheses and are clustered on nursing home; ***p < 0.01, **p < 0.05, *p < 0.1. Columns (4) and (6) use tickets related to system bugs as a dependent variable, and all other columns use end-user learning tickets as a dependent variable. *ORG* is the standardized sum of the five individual work practice dimensions, as described in the "Independent Variables" subsection.

All columns include month effects. Columns (5) through (7) also include facility effects. Columns (1) through (6) are Negative Binomial regressions and column (7) is a Poisson regression. Other results (not shown) are robust to removing outliers and to removing the first and last month. These results are available upon request.

significant ($t = 3.85$), which suggests that on average, nursing homes that use the work practices captured by the ORG measure tend to issue fewer support requests related to end-user learning, after controlling for other factors. In column (3), we interact nursing home experience with ORG. The negative, significant coefficient estimate on the interaction term indicates that after accounting for the main effects of experience and ORG, the interaction between the two is associated with lower numbers of support requests ($t = 2.0$), which is consistent with our key hypothesis that employees in high-ORG nursing homes learn more quickly as they use the technology. Because we use a nonlinear model, the magnitude of this interaction coefficient cannot be interpreted as in a linear regression (Karaca-Mandic et al., 2012). A marginal effects analysis of the estimates in equation (3) suggests that a one unit increase in nursing home experience is associated with 10% fewer support requests in nursing homes that are one standard deviation above the ORG mean than in homes that are one standard deviation below the ORG mean.

In column (4), as a robustness check, we report results when using system bugs as a dependent variable. The estimate on this term is not significantly different than zero, suggesting that the decline in this type of support request is not influenced by the work practices used by the nursing homes. The coefficient estimates on other variables are as expected, with support requests increasing with greater system usage by the nursing home and decreasing with greater vendor experience.

In columns (5) through (7), we include dummy variables for each nursing home. These account for many unobservable nursing home-level factors that might influence numbers of support requests issued by each nursing home and are not accounted for by our other control variables. The estimates in

column (5), which includes these variables, are consistent with the estimates from prior columns. With nursing home-level dummy variables, we still observe more rapid rates of decline in end-user learning requests in higher ORG nursing homes than in lower ORG nursing homes ($t = 2.50$).

In columns (6) and (7), we report estimates from two robustness tests that use the full specification including all controls and dummy variables for each nursing home and month. The first test uses support requests for system bugs as a dependent variable, and as in column (4), we find that our ORG measures are not correlated with the rate of decline in these bug requests. In column (7), we report results that use a Poisson rather than a negative binomial model, due to the consistency issues that arise with the negative binomial estimator described earlier. When using this estimator, the coefficient estimates are similar in sign and magnitude to those produced by the regressions using the negative binomial estimator ($t = 3.31$). In general, these results provide support for the argument that support requests reflecting end-user learning declined more rapidly in nursing homes that had higher ORG values.

Deconstructing the Organizational Measure

In Tables 4 and 5, we analyze the mechanisms through which these work practices might influence end-user learning by creating separate measures for each of the work practices that was aggregated to create the ORG measure. Our ability to separate the effects of the individual work practices in a regression framework is somewhat limited by having a relatively small number of observations and because these work practices tend to covary due to the fact that firms often bundle these practices together, as noted in the extensive literature on work systems. Nevertheless, the pattern of correlations can provide some insight into which specific work practices are most closely associated with learning outcomes.

We first report results from regressions that implement the model from Table 3 with dummy variables included for each nursing home, but we use the measures of specific work practices in columns (1) through (5), rather than using the aggregate ORG measure as in Table 3.[12] We find the strongest correlations between end-user learning and our measures of discretion and training. The training results are significant only at the 10% level. When we simultaneously include both the discretion and training measures in column (6), the estimates on training are no longer significant. These findings suggest that among the practices that form the ORG variable we analyzed in Table 3, discretion plays a particularly important role.

In Table 5, we use the occupational data in the surveys to further deconstruct each of the work practice measures. For each nursing home, we create three measures of each work practice by separating the survey responses from the major occupational groups within the nursing home: certified nursing assistants (CNA), nurses (NRS), and other allied professionals (ALP). Each of the subscripted measures in Table 5 indicates the value of that practice in the nursing home as measured using only the responses of workers who belong to a specific occupational category. This captures differences in the levels of discretion, for example, provided to workers in different occupations in the nursing home, and therefore allows us to test which of the occupational groups in the nursing home most benefit by being provided with greater discretion, in terms of their end-user learning outcomes.

When using the aggregate ORG measures in colums (1) and (2) of Table 5, we find that the decline in support request tickets is most closely associated with the responses of certified nursing assistants (i.e., CNAs), who tend to be low-skill frontline workers. This finding is consistent with prior work on IT and organizations that emphasizes frontline discretion as the central complement to IT investment. The estimates in columns (3) through (6) suggest that the discretion provided to CNAs, rather than their training, has a stronger effect on the rate at which end-user learning requests decline with experience ($t = 3.11$). As would be expected, the training of allied professionals, who normally enjoy relatively higher levels of discretion, appears to lower the overall number of support requests, although not by accelerating learning.[13] In sum, these estimates suggest that while training has the expected effects on reducing support requests for high-skill workers, a key differentiator across nursing homes in explaining end-user learning outcomes was the levels of discretion provided to frontline workers.

Discussion

Our analysis highlights the important role of work practices in explaining the nursing home's ability to absorb new IT systems. Our key finding is that structuring work in a manner

[12]We also have available upon request the results of additional tests that are based on independent coding of the support request ticket data into issue categories. Most importantly, the results from all of these robustness tests remain broadly consistent with the key arguments of the paper.

[13]For nurses, who already have decision authority and essentially act as managers in nursing homes, we found that it was most important to avoid problems with turnover or staffing shortages.

Table 4. Deconstructing the Organizational Measure into Specific Work Practices

Variables	(1)	(2)	(3)	(4)	(5)	(6)
Support Request Type	**End-User Learning**	**End-User Learning**	**End-User Learning**	**End-User Learning**	**End-User Learning**	**End-User Learning**
NH experience	0.016	0.011	0.000	0.002	-0.003	0.016
	(0.029)	(0.032)	(0.032)	(0.038)	(0.040)	(0.029)
DISC	1.554**					14.17**
	(0.666)					(5.658)
DISC × NH exp	-0.218**					-0.254*
	(0.106)					(0.140)
TEAM		1.501				
		(0.963)				
TEAM × NH exp		-0.191				
		(0.139)				
COMM			1.122			
			(1.065)			
COMM × NH exp			-0.103			
			(0.189)			
TRAIN				0.828**		-11.78**
				(0.388)		(5.104)
TRAIN × NH exp				-0.092*		0.051
				(0.056)		(0.118)
STAFF					0.606	
					(0.537)	
STAFF × NH exp					-0.042	
					(0.084)	
Vendor experience	-16.06**	-16.00**	-16.00**	-16.00**	-15.94**	-16.07**
	(7.133)	(7.029)	(6.950)	(7.064)	(6.953)	(7.098)
Month effects	Yes	Yes	Yes	Yes	Yes	Yes
Facility effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	185	185	185	185	185	185

Notes: ***p < 0.01, **p < 0.05, *p < 0.1. Robust standard errors are shown in parentheses and are clustered on nursing home; all regressions in Table 4 are negative binomial regressions and include month and facility fixed-effects. The dependent variable in all columns is the number of end-user learning tickets. All of the regressions in this table also include all of the control variables used in Table 3, column (5).

Column 1 uses measures of discretion (DISC). Column 2 uses measures of teamwork (TEAM). Column 3 uses measures of communication (COMM). Column 4 uses measures of training (TRAIN). Column 5 uses measures of staffing (STAFF). Column 6 includes measures of discretion (DISC) and training (TRAIN).

Table 5. Effects by Specific Work Practices and Occupation

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Support Request Type	End-User Learning	End-User Learning	End-User Learning	End-User Learning	End-User Learning	End-User Learning	End-User Learning	End-User Learning
NH experience	0.020	0.011	0.014	0.055**	-0.039	0.050	0.002	0.162***
	(0.044)	(0.052)	(0.045)	(0.026)	(0.056)	(0.064)	(0.044)	(0.037)
ORG_{CNA}	-0.176***	-0.003						
	(0.037)	(0.063)						
ORG_{NRS}	-0.041	0.410						
	(0.061)	(0.305)						
ORG_{ALP}	-0.390***	-0.097						
	(0.073)	(0.180)						
ORG_{CNA} × NH exp		-0.025**						
		(0.012)						
ORG_{NRS} × NH exp		-0.046						
		(0.046)						
ORG_{ALP} × NH exp		-0.006						
		(0.016)						
$DISC_{CNA}$			-0.344***	0.147			-0.856***	1.599*
			(0.075)	(0.166)			(0.306)	(0.890)
$DISC_{NRS}$			-0.102	2.056**			-0.235	2.125***
			(0.241)	(0.979)			(0.226)	(0.537)
$DISC_{ALP}$			-0.398***	-0.264**			0.692*	-1.907
			(0.088)	(0.119)			(0.405)	(1.991)
$DISC_{CNA}$ × NH exp				-0.066***				-0.318***
				(0.018)				(0.068)
$DISC_{NRS}$ × NH exp				-0.252				-0.264***
				(0.163)				(0.094)
$DISC_{ALP}$ × NH exp				0.007				0.158
				(0.016)				(0.129)
$TRAIN_{CNA}$					0.691	2.253*	0.066	3.549***
					(0.463)	(1.234)	(0.385)	(0.700)
$TRAIN_{NRS}$					-0.519	0.537	0.854	-3.317***
					(0.422)	(0.845)	(0.592)	(0.986)
$TRAIN_{ALP}$					-0.290***	-0.660**	-1.061***	1.183
					(0.080)	(0.277)	(0.381)	(1.809)
$TRAIN_{CNA}$ × NH exp						-0.080		-0.523***
						(0.147)		(0.088)
$TRAIN_{NRS}$ × NH exp						-0.004		0.512***
						(0.142)		(0.141)
$TRAIN_{ALP}$ × NH exp						-0.034		-0.167*
						(0.021)		(0.093)
Vendor experience	-19.43**	-19.16**	-19.75**	-19.26**	-19.07**	-19.38**	-19.74**	-20.81**
	(9.330)	(9.537)	(9.193)	(9.822)	(9.304)	(9.193)	(9.235)	(9.743)
Month effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Facility effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	134	134	134	134	134	134	134	134

Notes: Robust standard errors are shown in parentheses and are clustered on nursing home; ***p < 0.01, **p < 0.05, *p < 0.1. All regressions in Table 5 are negative binomial regressions and include facility fixed-effects. The dependent variable in all columns is the number of end-user learning tickets. All of the regressions include all of the control variables used in column (5) of Table 3, including for-profit status, log(beds), log(CNA minutes), and % of HIT use. DISC = Discretion, TRAIN = Training; The occupational subscripts on each measure are the standardized measure of that practice as reflected by the survey responses from that occupation. ALP = Allied Professionals; CNA = Certified Nursing Assistant, NRS = Nurses.

that enhances discretion, involvement, and skill can increase how quickly employees learn about new IT systems. IT implementation success, therefore, is contingent on client work practices, which suggests that vendors can anticipate the disruption that occurs during IT implementation if they account for the work structure of their clients. Moreover, regulators should consider coordinating incentive programs such as "meaningful use" requirements for IT adoption with organizational change initiatives, which, in the past, have largely involved separate efforts.

We also provide evidence for a specific channel through which complementary work practices influence IT-enabled frontline work. Prior work argues that providing discretion to frontline workers is advantageous because it facilitates rapid decision making, but our paper is the first to argue, to the best of our knowledge, that it also helps them to more rapidly learn about new technologies. From a practical standpoint, this suggests that, in healthcare, the return on investments in technology and training are likely to be greatest when accompanied by employee discretion. We also show that IT-complementary practices have an especially pronounced effect on low skilled (CNAs) and peripheral (allied professionals) occupational groups when compared to high skilled, core occupational groups (RNs).

As noted above, IT adoption and implementation have been far from consistent across healthcare organizations. One explanation for the mixed rate of adoption and implementation outcomes likely rests on the organizational infrastructure upon which new IT systems are introduced. Policy makers, vendors, and administrators pay close attention to the technological infrastructure necessary to meet the challenge of digitizing the U.S. healthcare system. Our findings serve as a reminder that alongside IT investment, the IT preparedness of the American healthcare system also rests on organizations' learning capabilities, which are a function of deliberate design, practices, and work arrangements.

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Lorin Hitt is currently the Zhang Jindong Professor of Operations, Information and Decisions at the University of Pennsylvania, Wharton School. His primary research and teaching is on the economics of information technology, industries heavily involved in the production or use of information technology (travel, banking, retail, healthcare, telecommunications, consumer products), and how changes in information affect consumer behavior, competition, and market structure. His recent work addresses how organizational complements affect the performance of IT investments, pricing strategies, and competition for information goods industries, the influence of recommender systems on consumer behavior, and the economic impact of social media. Lorin received his Sc.B. and Sc.M. degrees in Electrical Engineering from Brown University and his Ph.D. in Management from the Massachusetts Institute of Technology.